

# HelixCode — Open Issues Tracker

Per Constitution §11.4.15 (Item-status tracking) + §11.4.16 (Item-type tracking) + §11.4.19 (Fixed-document column-alignment) + CONST-057 (Type-aware closure vocabulary) + CONST-058 (Reopened-source attribution).

**Authoritative resumption ledger:** docs/CONTINUATION.md (CONST-044). This file complements it with item-level granularity for currently-open work.

**Status vocabulary** (closed set): Queued | In progress | Ready for testing | In testing | Reopened | Fixed/Implemented/Completed (→ Fixed.md)

**Type vocabulary** (closed set): Bug | Feature | Task

## Prefix convention

Round 189 (2026-05-19) introduced per-project / per-submodule ID prefixes replacing the legacy ISSUE-NNN flat namespace. New items **MUST** use the prefix matching their scope; cross-cutting items affecting two or more submodules (or any meta-repo concern) live under HXC. Numeric portion is per-prefix (each prefix starts at 001). Legacy ISSUE-NNN IDs renamed forward-only per CONST-043; historical close-out narratives in docs/CONTINUATION.md preserve original IDs verbatim, and docs/Fixed.md annotates each closure with (ex-ISSUE-NNN) for git-history traceability.

| Prefix | Scope                                                                              | Source                                                 |
|--------|------------------------------------------------------------------------------------|--------------------------------------------------------|
| HXC    | HelixCode root project (project-wide, multi-submodule, governance, infrastructure) | this repo                                              |
| HXA    | HelixAgent submodule                                                               | dependencies/<br>HelixDevelopment/<br>HelixAgent (when |

| Prefix | Scope                                                    | Source                                                    |
|--------|----------------------------------------------------------|-----------------------------------------------------------|
|        |                                                          | present) /<br>helix_agent/ tree                           |
| HXM    | HelixMemory<br>submodule                                 | dependencies/<br>HelixDevelopment/<br>HelixMemory         |
| HXL    | HelixLLM<br>submodule                                    | dependencies/<br>HelixDevelopment/<br>HelixLLM            |
| HXQ    | HelixQA submodule                                        | dependencies/<br>HelixDevelopment/<br>HelixQA (helix_qa/) |
| HXS    | HelixSpecifier<br>submodule                              | dependencies/<br>HelixDevelopment/<br>HelixSpecifier      |
| HXO    | HelixOrchestrator<br>(=<br>LLMOrchestrator)<br>submodule | dependencies/<br>HelixDevelopment/<br>LLMOrchestrator     |
| HXV    | HelixVerifier (=<br>LLMsVerifier)<br>submodule           | dependencies/<br>HelixDevelopment/<br>LLMsVerifier        |
| HXD    | HelixDocProcessor<br>(= DocProcessor)<br>submodule       | dependencies/<br>HelixDevelopment/<br>DocProcessor        |
| HXI    | HelixI18n (when<br>added)                                | tba                                                       |
| PLN    | Planning<br>submodule (vasic-<br>digital)                | dependencies/vasic-<br>digital/Planning                   |
| VEN    | VisionEngine<br>submodule<br>(HelixDevelopment)          | dependencies/<br>HelixDevelopment/<br>VisionEngine        |
| SLF    | SelfImprove<br>submodule (vasic-<br>digital)             | dependencies/vasic-<br>digital/SelfImprove                |
| STO    | Storage submodule<br>(vasic-digital)                     | dependencies/vasic-<br>digital/Storage                    |
| OPS    | LLMOps<br>submodule (vasic-<br>digital)                  | dependencies/vasic-<br>digital/LLMOps                     |
| VDB    | VectorDB<br>submodule (vasic-<br>digital)                | dependencies/vasic-<br>digital/VectorDB                   |
| OBS    |                                                          |                                                           |

| Prefix | Scope                                   | Source                                    |
|--------|-----------------------------------------|-------------------------------------------|
|        | Observability submodule (vasic-digital) | dependencies/vasic-digital/Observability  |
| MCP    | MCP_Module submodule (vasic-digital)    | dependencies/vasic-digital/MCP_Module     |
| MSG    | Messaging submodule (vasic-digital)     | dependencies/vasic-digital/Messaging      |
| MDW    | Middleware submodule (vasic-digital)    | dependencies/vasic-digital/Middleware     |
| PLG    | Plugins submodule (vasic-digital)       | dependencies/vasic-digital/Plugins        |
| STR    | Streaming submodule (vasic-digital)     | dependencies/vasic-digital/Streaming      |
| WAT    | Watcher submodule (vasic-digital)       | dependencies/vasic-digital/Watcher        |
| CNV    | conversation submodule (vasic-digital)  | dependencies/vasic-digital/conversation   |
| AUT    | Auth submodule (vasic-digital)          | dependencies/vasic-digital/Auth           |
| LZY    | Lazy submodule (vasic-digital)          | dependencies/vasic-digital/Lazy           |
| ATP    | AutoTemp submodule (vasic-digital)      | dependencies/vasic-digital/auto_temp      |
| PLI    | PliniusCommon submodule (vasic-digital) | dependencies/vasic-digital/plinius_common |
| CHL    | challenges submodule (vasic-digital)    | challenges/                               |
| CNT    | containers submodule                    | containers/                               |
| SEC    | security submodule                      | security/                                 |
| PAN    | panoptic submodule                      | panoptic/                                 |

For submodules not listed above, default to the first 3 letters of the submodule name, uppercase (e.g. Watcher → WAT). Document the new prefix in this table on first use.

## Legacy → new mapping (round 189)

| Old ID    | New ID  | Scope rationale                                               |
|-----------|---------|---------------------------------------------------------------|
| ISSUE-001 | VEN-001 | VisionEngine helix-gitlab URL                                 |
| ISSUE-002 | VEN-002 | VisionEngine vasic-digital-github fork divergent              |
| ISSUE-003 | HXL-001 | HelixLLM analysis_test.go hardcoded path                      |
| ISSUE-004 | HXL-002 | HelixLLM TOON WriteT00N 500                                   |
| ISSUE-005 | HXC-001 | CONST-052 rename programme (project-wide)                     |
| ISSUE-006 | HXC-002 | Round-74 residual LOGIC FAILs (multi-submodule cross-cutting) |
| ISSUE-007 | HXC-003 | CONST-046 migration backlog (project-wide)                    |
| ISSUE-008 | HXQ-001 | helix_qa TestPerformance flake                                |
| ISSUE-009 | HXA-001 | helix_agent handler tests (4)                                 |
| ISSUE-010 | HXA-002 | helix_agent debate API drift                                  |
| ISSUE-011 | HXA-003 | venice CONST-037 (in helix_agent)                             |

## VEN-001 (ex-ISSUE-001) — VisionEngine helix-gitlab remote repo missing (404) — CLOSED (→ Fixed.md)

**Status:** Completed (→ Fixed.md) **Type:** Task **Discovered:** 2026-05-19 (round 98 — Planning + VisionEngine i18n migration) **Discovered-By:** AI subagent during 4-remote push attempt **Closed-By:** Round 188 (subagent repo-inventory sweep) **Root cause:** The helix-gitlab remote URL in dependencies/HelixDevelopment/VisionEngine/.git/config pointed at git@gitlab.com:HelixDevelopment/visionengine.git — a non-existent group path. The actual GitLab group is helixdevelopment1 (path) / HelixDevelopment (display name). The repository helixdevelopment1/VisionEngine (id 80411994) already existed since 2026-03-19. NOT a missing-repo issue — a URL-misconfiguration issue. **Fix:** git remote set-url helix-gitlab git@gitlab.com:helixdevelopment1/VisionEngine.git in the VisionEngine submodule, then git push helix-gitlab master (FF-safe: local was 46 commits ahead, remote 0 ahead). Push landed at SHA 2d0c35b (verified via git ls-remote helix-gitlab master). The Upstreams recipe push-helix-gitlab.sh references the remote by name (not URL), so it continues to work unchanged. **Evidence:** - glab api projects/helixdevelopment1%2Fvisionengine → id 80411994 OK -

git ls-remote helix-gitlab HEAD →  
2d0c35bebb199a9a199fbf899eaeb292e38eaf17 (matches local HEAD) -  
Original broken URL still 404s when probed directly (proves URL  
was the issue, not perms)

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## **HXL-001 (ex-ISSUE-003) — HelixLLM internal/agents/tools/analysis\_test.go hardcoded absolute path**

**Status:** Fixed (→ Fixed.md) **Type:** Bug **Discovered:** 2026-05-19  
(round 95 — HelixLLM migration; surfaced as pre-existing failure)  
**Discovered-By:** AI subagent during HelixLLM standalone test  
run **Closed-By:** Round 105 (commit a5e56d4 in HelixLLM; meta  
pointer fedd152) **Attribution correction:** Originally documented  
as helix\_agent; actual location is HelixLLM submodule  
(dependencies/HelixDevelopment/HelixLLM/internal/agents/tools/).  
Commit SHAs 0a84310 resolved there. **Resolution:** Replaced  
hardcoded path with t.TempDir() + 2 synthesised fixture files.  
Bonus: same bug-pattern discovered in git\_test.go (constant  
helixLLMRoot + 7 tests) — refactored gitSandbox() signature. 6 tests  
now PASS on any host. Mutation verified.

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## **HXL-002 (ex-ISSUE-004) — HelixLLM internal/gateway/middleware TOON WriteT00N returns 500**

**Status:** Fixed (→ Fixed.md) **Type:** Bug **Discovered:** 2026-05-19  
(round 95) **Discovered-By:** AI subagent **Closed-By:** Round 105  
(commit a5e56d4) **Attribution correction:** Originally documented  
as helix\_agent; actual location is HelixLLM submodule. Commit  
6f11c56 resolved there. **Resolution:** Root cause was vasic-digital/  
TOON's round-27 anti-bluff change (Marshal returns  
ErrT00NEncodingNotImplemented unconditionally) combined with  
WriteT00N treating ANY Marshal error as 500. Fix: fall back to  
json.Marshal while preserving application/toon Content-Type  
(matches ContentNegotiation middleware). 500 still returned for  
genuinely unmarshallable values (channels). 19 middleware tests  
now PASS. Mutation verified.

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# HXC-001 (ex-ISSUE-005) — CONST-052

## rename programme: meta-repo

## directories still PascalCase

**Status:** In progress **Type:** Task **Discovered:** 2026-05-15 (CONST-052 cascade landed) **Discovered-By:** Constitution **Evidence:** Meta-repo top-level dirs already snake\_case (round-88 sweep). Remaining non-compliance is two layers deeper: dependencies/HelixDevelopment/\* + dependencies/vasic-digital/\* owned-org submodule dirs (PascalCase), and 59 Upstreams/ dirs inside submodule trees. **Resolution path:** Phased migration per CONST-052 §11.4.29. Round 113 produced the phased plan (f666410, docs/superpowers/specs/2026-05-19-const052-rename-programme-plan.md). Round 343 executed the safe (zero-submodule-go.mod-entanglement) batches.

**Round-343 12 chosen snake\_case names (operator “agent defaults”):** D-1 sequential phases; D-2 helix\_development (parent dir, deferred — touches every consumer go.mod); D-3 vasic-digital kept (GitHub-org handle, proper-noun carve-out); D-4 n/a (helix\_code already snake\_case); D-5 LLMsVerifier/Assets+Website deferred (deployment-wire audit); D-6 mcp\_module; D-7 i\_llm; D-8 toon; D-9 rag; D-10 cluster-C upstreams strict; D-11 yes co-authored; D-12 one approval per batch.

### Round-343 per-batch status:

| Batch | Renamed                                                                                                                                                                                                                                                                                              | Status            | Evidence                                                                   |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|----------------------------------------------------------------------------|
| 1     | HelixDevelopment/<br>Models → models                                                                                                                                                                                                                                                                 | LANDED<br>alea3c8 | submodule<br>resolves; go<br>build ./<br>internal/... ./<br>cmd/... exit 0 |
| 2     | HelixDevelopment/<br>DebateOrchestrator<br>→<br>debate_orchestrator<br><br>11 vasic-digital/*<br>zero-go.mod-<br>consumer leaves<br>(auto_temp, claritas,<br>doc_processor,<br>gandalf_solutions,<br>hyper_tune, i_llm,<br>leak_hub,<br>ouroborous,<br>plinius_common,<br>veritas,<br>vision_engine) | LANDED<br>416fe8e | go list -m<br>digital.vasic.debate<br>→ new path; build<br>exit 0          |
| 3     |                                                                                                                                                                                                                                                                                                      | LANDED<br>e813b5c | 11 submodule<br>statuses resolve;<br>build exit 0                          |

| Batch    | Renamed                                                                                         | Status   | Evidence                                                                                                                                                         |
|----------|-------------------------------------------------------------------------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Deferred | ~37 owned-org<br>leaves consumed<br>by helix_agent/<br>helix_qa/HelixLLM<br>go.mod              | DEFERRED | renaming requires<br>submodule-internal<br>go.mod commits<br>entangled with pre-<br>existing<br>uncommitted work<br>— needs dedicated<br>per-submodule<br>rounds |
| Deferred | parent dirs<br>HelixDevelopment/<br>→helix_development/<br>(D-2), vasic-<br>digital/ kept (D-3) | DEFERRED | parent rename<br>touches every<br>consumer go.mod<br>atomically                                                                                                  |
| Deferred | 59 Upstreams/<br>→upstreams/<br>(cluster C, D-10)                                               | DEFERRED | live inside<br>submodule trees —<br>separate-repo<br>commits                                                                                                     |

13 of ~58 owned-org leaf renames done this round (zero build breakage). HXC-001 stays In progress pending the deferred submodule-entangled and parent-dir batches.

## HXC-002 (ex-ISSUE-006) — Round-74 residual LOGIC-class FAILs (CLOSED)

**Status:** Fixed (→ Fixed.md) **Type:** Bug **Discovered:** 2026-05-19 (round 74 — release-gate-test.sh creation; classified by round 89) **Discovered-By:** AI release-gate sweep **Closure progress:** - ✓ **HelixMemory:** closed round 106 (commit 69016df — single-line go.mod fix; 6 FAIL → 0 FAIL) - ✓ **vasic-digital/Planning:** round 107 NO-OP — 275 PASS / 0 FAIL / 20 SKIP-OK; likely incidentally fixed by round 98 i18n migration - ✓ **helix\_agent inner:** closed round 109 (commit 0f492e98 — 5 test-side bluff fixes, zero production changes) **Evidence:** Round 74 surfaced 26 FAILs across submodules; rounds 82-87 closed 19; this Issue tracked the residual 7 across 3 submodules. All 3 components closed by rounds 106 + 107 + 109. **Follow-ups surfaced (NEW issues filed):** 4 helix\_agent handler tests previously masked by mid-run panic (now visible) + 3 build-failed packages depending on sibling submodule API drift (digital.vasic.debate) + 2 LOGIC FAILs reclassified as cross-cutting work (venice CONST-037 model-wiring + compliance CONST-051 architectural reconciliation). See HXA-001 through HXA-003 (filed below).

## HXA-001 (ex-ISSUE-009) — helix\_agent handler tests surfaced after round-109 fix

**Status:** Queued **Type:** Bug **Discovered:** 2026-05-19 (round 109)  
**Discovered-By:** AI subagent (helix\_agent LOGIC audit)  
**Evidence:** Mid-run panic in TestIsProviderAvailable\_NotAvailable aborted test binary; round 109's fix unblocked execution, surfacing 4 pre-existing FAILs:  
TestFormattersHandler\_FormatCode\_UnsupportedLanguage,  
TestEmbeddingHandler\_WithRealManager, TestGetTaskResources,  
TestGetTaskLogs. Out of round-109's 5-fix cap. **Resolution path:**  
Per-handler investigation, similar to round 109's test-side bluff pattern.

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## HXA-002 (ex-ISSUE-010) — helix\_agent debate/llmprovider sibling-submodule API drift

**Status:** Fixed (→ Fixed.md) **Type:** Bug **Discovered:** 2026-05-19 (round 109) **Discovered-By:** AI subagent **Investigated:** 2026-05-20 (round 324) — split into a mechanical part and a design-decision part. **Closed:** 2026-05-20 (round 342) **Closure-Ref:** helix\_agent commit (round-342 HXA-002 debate API drift) + meta-repo .gitmodules pointer-bump **Investigation finding (operator's explicit ask — moved vs deleted):** The learning/knowledge/recommendations capability tier was **genuinely DELETED, not moved**. git log on the digital.vasic.debate submodule (dependencies/HelixDevelopment/debate\_orchestrator, renamed from DebateOrchestrator per CONST-052) shows the orchestrator was rebuilt from scratch — commit 196d0ea "feat: initial DebateOrchestrator reconstruction (Phase 1)". orchestrator/api.go has carried only the slim CreateDebate/GetStatistics surface since that very first commit (git log --follow orchestrator/api.go = single entry). A tree-wide grep of dependencies/ for KnowledgeRepository, GetRecommendations, ConvertAPIRequest, GetDebateStatus, DefaultMinConsensus, MaxAgentsPerDebate, EnableAgentDiversity found **zero** surviving copies in any digital.vasic.\* package or in HelixSpecifier/HelixMemory. The slim API is the first and only version — the richer tier was a pre-reconstruction artifact that no longer exists anywhere. Per the operator's chosen direction for the deleted case, the helix\_agent tests were rewritten down to the slim API. **Resolution:** See docs/Fixed.md row for the full closure narrative (Part-1 import swap + Part-2 slim-API rewrite + score-scale + go.mod rename-drift fix + captured evidence).

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## HXA-003 (ex-ISSUE-011) — venice TestGetCapabilities model-list drift (CONST-037)

**Status:** Fixed (→ Fixed.md) **Type:** Bug **Discovered:** 2026-05-19 (round 109) **Discovered-By:** AI subagent **Closed:** 2026-05-19 (round 190) **Closure-Ref:** helix\_agent commit (round-190 venice CONST-037 model-list drift) + meta-repo pointer-bump **Evidence:** Test hardcoded venice-uncensored; Venice API returned 75 models with the family rotated to venice-uncensored-1-2 / venice-uncensored-role-play. Per CONST-037 (LLMsVerifier is the single source of truth for model metadata) the assertion violated the no-hardcoded-list rule. **Resolution:** helix\_agent/internal/llm/providers/venice/venice\_test.go::TestGetCapabilities — replaced `assert.Contains(..., "venice-uncensored")` and `assert.Contains(..., "llama-3.3-70b")` with structural assertion: `NotEmpty(SupportedModels)` plus a substring scan for the `venice-uncensored*` family. SKIP-OK marker per CONST-035 fires if the entire family disappears (avoids false-positive PASS). Mutation-verified (revert → FAIL with the original drift, restore → PASS).

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## HXC-004 — Recovery-batch under- verification (40% FAIL rate per round-193 audit)

**Status:** Fixed (→ Fixed.md) **Type:** Bug **Discovered:** 2026-05-19 (round 193 — recovery-batch verification audit) **Discovered-By:** AI subagent **Fixed:** 2026-05-19 (round 200 — per-package test-assertion repair) **Evidence:** Round-193 audit of 10 recovery-batch-landed packages (recovery commits b7f8672 + 5c94696) found 6 PASS / 4 FAIL: - internal/llm (round 161): test-assertion drift — tests still expected pre-i18n English literal “api\_key”, production emits message-ID `internal_llm_wizard_anthropic_apikey_required` - internal/logo (round 163): same drift — tests expected “failed to open” / “failed to decode”, production emits `internal_logo_open_source_failed` / `internal_logo_decode_source_failed` - internal/notification (round 167): same drift — tests expected Title literals “Task Completed”, “Task Failed”, “Workflow Completed”, “Workflow Failed”, “Worker Disconnected”, “System Error”, “System Started”; production emits `internal_notification_title_*` IDs - internal/performance (round 168): build break — `translator.go` `stdctx.Context` vs plain context — fixed inline by parent agent **Root cause:** Recovery-batch commits captured stalled-agent file content but did NOT re-run consuming-test updates + did NOT verify build/test green per-package. **Resolution:** Round-200 subagent updated test assertions in all 3 drifted packages to expect message-ID echoes

(internal\_<pkg>\_\* prefix). Per-package PASS confirmed (llm: 51.8s, logo: 0.07s, notification: 0.89s, performance: 8.4s). Per CONST-035 mutation-verified one assertion per package: revert to literal → FAIL (production emits the ID), restore → PASS. **Audit reference:** docs/audits/2026-05-19-recovery-batch-verification.md (commit 1badef1).

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## HXC-003 (ex-ISSUE-007) — CONST-046 migration backlog (57,329 violations baselined; shrinking)

**Status:** In progress **Type:** Feature **Discovered:** 2026-05-19 (round 92 — audit script) **Discovered-By:** AI subagent ground-truth scan **Evidence:** Round-92 scan reported 57,345 violations across 21,937 files. Round 99b baseline collapsed to 54,803 unique (path, literal\_hash) keys. Phase 4 (rounds 100+) systematically migrating top-concentration files: round 100 (evaluators.go), 101 (challenge\_recorded\_ai testgen.go), 102 (challenge\_desktop.go) — see CONTINUATION.md close-outs. **Resolution path:** Continued Phase 4 cadence; audit-gate --fail-on-new already enforced; each migration round MUST re-run --update-baseline so snapshot shrinks toward zero.

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## HXQ-001 (ex-ISSUE-008) — helix\_qa intermittent TestPerformance flake (host-load-sensitive)

**Status:** Fixed (→ Fixed.md) **Type:** Bug **Discovered:** 2026-05-19 (round 82) **Discovered-By:** AI subagent **Fixed:** 2026-05-20 (round 325) **Evidence:** helix\_qa TestPerformance (three perf tests in pkg/vision/ — TestPerformance\_DHash64\_Under5msPer1080pFrame, TestPerformance\_PHash\_Under25msPer1080pFrame, TestPerformance\_SSIM\_Under5msPer480pFrame) fails intermittently under high host load (concurrent containers + builds). Not a code bug per se; a sensitivity issue — the hard per-frame timing ceilings (5 ms / 25 ms / 5 ms) are only meaningful on a quiescent host. **Resolution:** Decision — **path (b)** (env-var gating) chosen over path (a) (loosen tolerance). Rationale: loosening the timing tolerance would weaken the test's anti-bluff value — a genuine perf regression could then pass. Path (b) preserves the strict assertions while making the flake deterministic: the three tests now check os.Getenv("HOST\_LOAD\_DEDICATED") and t.Skip("SKIP-OK: #HXQ-001 ...") honestly when unset, running strict only on a quiescent dedicated host (HOST\_LOAD\_DEDICATED=1). This is the CONST-035-compliant choice. Landed in helix\_qa submodule

commit 649e2dd + meta .gitmodules pointer bump. docs/test-coverage.md §6.1 documents the env var. Post-fix evidence: go build ./pkg/vision/... exit 0, go vet clean; go test -count=1 -run TestPerformance ./pkg/vision/... (unset) → all 3 --- SKIP with SKIP-OK: #HXQ-001 marker; HOST\_LOAD\_DEDICATED=1 go test -count=1 -run TestPerformance ./pkg/vision/... → all 3 --- PASS strict (DHash64 average 741ns, PHash average 88.969µs — well under the 5 ms / 25 ms ceilings).

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## HXC-005 — cmd/ performance\_optimization\_standalone/main.go is a CONST-035 simulation bluff

**Status:** Fixed (→ Fixed.md) **Type:** Bug **Discovered:** 2026-05-20 (round 317 — cmd i18n migration subagent) **Discovered-By:** AI subagent — refused to localize the file because doing so would polish a bluff **Fixed:** 2026-05-20 (round 318) **Evidence:** helix\_code/cmd/performance\_optimization\_standalone/main.go was a package main that printed “ Starting HelixCode Production Performance Optimization” then *simulated* every optimization phase: // Simulate production optimization phases, time.Sleep(500 \* time.Millisecond) per phase, and improvement := 5.0 + rand.Float64()\*20.0 — fabricated improvement percentages from a random number generator. No real profiling, no real optimization, no real measurement. The canonical BLUFF-001-class anti-pattern (CLAUDE.md §3.3 / §6 ANTI-PATTERN 1) — a binary that reports success for work it never performed. Violated CONST-035 / Article XI §11.9. **Resolution:** Decision — **DELETE** (resolution path b). The standalone tool was genuinely obsolete: fully superseded by cmd/performance\_optimization/ (snake\_case post-CONST-052), which calls the REAL dev.helix.code/internal/performance.PerformanceOptimizer — real runtime.ReadMemStats, real GOMAXPROCS tuning, real before/after measurement — and carries CONST-046 i18n + a unit-test file. git rm -r cmd/performance\_optimization\_standalone/ removed the dead bluff; stale references purged from docs/COMPREHENSIVE\_AUDIT\_REPORT.md. Reproduce-before-fix Challenge added at cmd/performance\_optimization/bluff\_regression\_test.go: TestHXC005\_BluffStandaloneDirectoryDeleted asserts the obsolete path is gone + the real command survives; TestHXC005\_RealOptimizerMeasuresActualMemory allocates a retained 32 MiB buffer and asserts the optimizer’s baseline MemoryUsage tracks a genuine runtime.HeapAlloc reading (not an RNG single-digit). Post-fix evidence: go build ./cmd/... exit 0; go test -count=1 -run TestHXC005 ./cmd/performance\_optimization/ → both PASS (literal log: optimizer baseline MemoryUsage=33812624 bytes, runtime.HeapAlloc=33802008 bytes — both real measurements,

same order of magnitude. No RNG-fabricated improvement percentages.); anti-bluff smoke on the deleted path returns N/A (directory gone).

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## **PAN-001 — panoptic appendString truncates multi-byte UTF-8 runes to bytes (TestResult.MarshalJSON corrupts non-ASCII)**

**Status:** Fixed (→ Fixed.md) **Type:** Bug **Discovered:** 2026-05-19 (round 298 — panoptic enrichment subagent) **Discovered-By:** AI subagent runner-detector against real executor.TestResult.MarshalJSON **Fixed:** 2026-05-19 (round 302 — panoptic submodule commit 24aa627 + meta pointer bump) **Evidence:** panoptic/internal/executor/executor.go:120 — buf = append(buf, byte(r)) in the else branch of appendString casts a rune to a single byte. Multi-byte UTF-8 codepoints (German umlauts, Spanish accents, Japanese CJK, Serbian Cyrillic, Chinese Han) get truncated to one byte each, producing corrupted JSON output. Honestly tracked via the round-298 Challenge runner's executor-marshal:utf8-detector:regression-present PASS line + KNOWN-ISSUE entry in panoptic/docs/test-coverage.md §7. Affects every consumer that JSON-marshals a TestResult containing non-ASCII text. **Resolution:** Replaced buf = append(buf, byte(r)) with buf = utf8.AppendRune(buf, r) (Go 1.21+) + added unicode/utf8 import. Single-line functional fix. Post-fix evidence: go test -race -count=1 ./internal/executor/... → ok 4.470s; bash challenges/panoptic\_describe\_challenge.sh → 39/39 PASS, 0 FAIL; runner UTF-8 detector flipped from regression-present → fixed (literal log: PASS [executor-marshal:utf8-detector:fixed] + KNOWN-ISSUE-RESOLVED: executor.appendJSONString now UTF-8 clean). Closed in this round.

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## **HXQ-002 — helix\_qa pkg/autonomous ↔ VisionEngine remote API drift blocks helix\_agent tests/integration compile**

**Status:** Queued **Type:** Bug **Discovered:** 2026-05-20 (round 342 — HXA-002 resolution subagent) **Discovered-By:** AI subagent — surfaced as a compile failure separate from HXA-002 while verifying the rewritten helix\_agent integration tests **Evidence:** helix\_agent's tests/integration package cannot compile because helix\_qa/pkg/autonomous (committed helix\_qa code at 7fa674a) consumes a VisionEngine remote package API that has since

drifted. The failure is entirely unrelated to HXA-002's debate-API drift — it is a distinct second drift between the helix\_qa autonomous-session package and the VisionEngine remote package. Round-342 left it untouched per strict scope.

**Resolution path:** Dedicated round — diff the VisionEngine remote package's current exported surface against what helix\_qa/pkg/autonomous (at or after 7fa674a) imports; classify each mismatch (renamed symbol / changed signature / removed type). If mechanical, update the helix\_qa/pkg/autonomous consumer to the current remote API (note: round-340 VEN-002 merge-first made VisionEngine remote a strict superset — the EnsureReady SSH-probe variant — so the drift may now be resolvable by consuming the merged superset). Verify helix\_agent tests/integration compiles + passes afterward.

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*Last regenerated: 2026-05-20 (round 342 — HXQ-002 added). To update Issues\_Summary.md mechanically, run scripts/generate\_issues\_summary.sh (TODO: create — currently this Issues.md is the source of truth and Summary is hand-maintained).*